

M. UMER MAALIK

Mechanical Engineering & Physics

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OBJECTIVE

Highly motivated Mechanical Engineering and Physics student with a strong foundation in first-principles problem solving. Experienced in electro-mechanical design, structural analysis (FEA), and thermal/fluid dynamics (CFD) using Ansys.

EDUCATION

B.S. in Mechanical Engineering & Physics

Stony Brook University, Stony Brook, NY

August 2025 – May 2030

EXPERIENCE

East End Assemblies

June 2026 – Present

Mechanical Engineering Intern

Long Island, NY

- Designed and developed mechanical components and assemblies using Fusion 360 and SolidWorks, contributing to production-ready designs from concept through detailed modeling.
- Collaborated with the engineering team to iterate on part designs, ensuring manufacturability and adherence to project specifications.

Stony Brook Robotics

September 2025 – Present

FEA Lead

Stony Brook, NY

- Spearheaded structural (FEA) and fluid (CFD) analysis using Ansys Mechanical and Fluent to optimize the MATE Rover's chassis for deep-water pressures and extreme stresses.
- Saved critical time and resources by providing predictive data on component performance under extreme stress and atmospheric conditions.
- Optimized the MATE Rover's hydrodynamic profile using Ansys Fluent (CFD), significantly reducing drag and increasing efficiency for underwater environments.

Autodesk, Stony Brook

January 2026 – Present

Autodesk Ambassador

Stony Brook, NY

- Hosted an interdisciplinary workshop instructing Mechanical, Electrical, and Arts students in advanced Fusion 360 tools, finite element analysis (FEA), and rendering techniques.

PROJECTS

iU Cosmic Gamma-Ray Burst Nanosatellite

July 2026 – Present

Solidworks, EasyEDA, System Architecture, Space Systems, NASA CSLI

- Founded the LEO-GRM nanosatellite initiative, assembled the project team, secured faculty advisory support, and authored the technical proposal targeting NASA's CubeSat Launch Initiative (CSLI).
- Architected a $100 \times 100 \times 113.5$ mm iU CubeSat incorporating a 6-stage vertical PCB stack, CsI(Tl) scintillation crystal, and SiPM array for soft gamma-ray burst detection.
- Engineered the structural chassis envelope in Fusion 360 using solid Aluminum 6061-T6 to meet NASA Launch Services Program vibration, mass (1.33 kg), and spatial constraints.
- Managed project regulatory compliance, coordinating IARU frequency allocation (435–438 MHz), FCC Part 5 licensing (Form 442), and DAS orbital debris burn-up analysis.

Fully Autonomous 3D Printed Jet

December 2025 – Present

Fusion 360, Inventor, Ansys Mechanical, Ansys Fluent, Neural Networks

- Engineered a fully customized jet using Fusion 360 and Inventor capable of speeds exceeding 70 km/h, integrating a thrust vectoring mechanism for hard cutting maneuvers.
- Designing and training a neural network for fully autonomous flight control and landing.
- Analyzed aerodynamics using Ansys Mechanical and Ansys Fluent (CFD) to optimize structural integrity and resistance to turbulence.

Thrust Vector Control (TVC) System

Fusion 360, Open Rocket

- Developed a TVC system to precisely manage rocket trajectory by adjusting engine position to counteract air currents.
- Designed and integrated complex electro-mechanical assemblies in Fusion 360, successfully packaging motors, microcontrollers, and PCBAs into tight aerospace tolerances.
- Engineered mounting systems to mitigate shock and vibration, ensuring the structural integrity and reliability of onboard electronics during high thrust operation.
- Engineered the system for full compatibility with a 20×4 inch rocket weighing approximately 200 grams.

SKILLS

CAD / PLM

Siemens NX, SolidWorks, Fusion 360, Inventor, Creo, Onshape, Solid Edge

Analysis

Ansys Mechanical (FEA), Ansys Fluent (CFD), Ansys Workbench, LS-DYNA, OpenFOAM (Beginner), SimScale

Software

Python, KiCAD (PCBAs), PyTorch, Qiskit, Open Rocket, L^AT_EX (Overleaf), Microsoft Office